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Contents

List of Abbreviations	ii
1 Introduction	1
2 LTE-Advanced General Architecture	2
3 User Equipment (UE)	4
3.1 Mobile Equipment (ME)	4
3.2 Universal Subscriber Identity Module (USIM)	4
4 Radio Access Network (E-UTRAN)	5
4.1 Evolved NodeB (eNodeB)	5
5 Evolved Packet Core (EPC)	7
5.1 Mobility Management Entity (MME)	7
5.2 Home Subscriber Server (HSS)	7
5.3 Serving Gateway (S-GW)	8
5.4 PDN Gateway (P-GW)	8
5.5 Policy and Charging Rules Function (PCRF)	8
6 LTE-Advanced Protocol Stack	10
6.1 User Plane Protocol Layers	10
6.2 Control Plane: RRC Layer	11
6.3 Non-Access Stratum (NAS)	11
7 Conclusion	12
References	13

List of Figures

1	LTE-Advanced (EPS) Overall Architecture and Key Interfaces [4]	2
2	Internal Structure of the E-UTRAN and Interfaces with EPC and ME [3] . . .	5
3	Evolved Packet Core (EPC) Detailed Architecture with Interfaces [3]	7

List of Tables

1	List of Common Abbreviations used in LTE-Advanced Architecture	ii
2	Key Entities of the EPS Architecture and their Primary Functions	2

List of Abbreviations

Table 1: List of Common Abbreviations used in LTE-Advanced Architecture

Abbreviation	Definition
3GPP	3rd Generation Partnership Project
4G	4th Generation
5G	5th Generation
APN	Access Point Name
AMBR	Aggregate Maximum Bit Rate
ARP	Allocation and Retention Priority
AuC	Authentication Center
DPI	Deep Packet Inspection
eNB	Evolved NodeB
EPC	Evolved Packet Core
EPS	Evolved Packet System
E-UTRAN	Evolved Universal Terrestrial Radio Access Network
GTP	GPRS Tunnelling Protocol
HARQ	Hybrid Automatic Repeat Request
HLR	Home Location Register
HSS	Home Subscriber Server
IMSI	International Mobile Subscriber Identity
IP	Internet Protocol
LTE	Long Term Evolution
LTE-A	LTE-Advanced
MAC	Medium Access Control
ME	Mobile Equipment
MIMO	Multiple-Input Multiple-Output
MME	Mobility Management Entity
NAS	Non-Access Stratum
OFDMA	Orthogonal Frequency Division Multiple Access
PAPR	Peak-to-Average Power Ratio
PCC	Policy and Charging Control
PCEF	Policy and Charging Enforcement Function
PCRF	Policy and Charging Rules Function
PDCCP	Packet Data Convergence Protocol

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Table 1 – continued from previous page

Abbreviation	Definition
PDN	Packet Data Network
PHY	Physical Layer
P-GW	PDN Gateway
QCI	QoS Class Indicator
QoS	Quality of Service
RLC	Radio Link Control
ROHC	Robust Header Compression
RRC	Radio Resource Control
RRM	Radio Resource Management
SAE	System Architecture Evolution
SC-FDMA	Single Carrier Frequency Division Multiple Access
S-GW	Serving Gateway
SON	Self-Organizing Networks
TA	Tracking Area
UE	User Equipment
USIM	Universal Subscriber Identity Module
X2	Interface between eNodeBs

Abstract

This report presents a comprehensive structural analysis of the Long Term Evolution-Advanced (LTE-A) architecture, standardized by the 3rd Generation Partnership Project (3GPP) as the Evolved Packet System (EPS). Designed to meet IMT-Advanced requirements [1], the EPS introduces a flat, all-IP framework that supersedes legacy centralized network models. The architecture is systematically partitioned into three primary domains: the User Equipment (UE), the Evolved Universal Terrestrial Radio Access Network (E-UTRAN), and the Evolved Packet Core (EPC). The analysis highlights the distribution of radio resource management directly to the evolved NodeBs (eNodeBs) and details the strict segregation of control and user planes within the EPC. Furthermore, the report examines the essential core network entities (MME, HSS, S-GW, P-GW, PCRF) and dissects the layered protocol stack across both the Access Stratum (AS) and Non-Access Stratum (NAS). Ultimately, this evaluation demonstrates how LTE-A's structural principles established the foundational design paradigms for modern and next-generation cellular networks [2].

1. Introduction

The relentless surge in global mobile data traffic necessitated a fundamental re-architecture of cellular networks. In direct response, the **3rd Generation Partnership Project (3GPP)** developed **Long Term Evolution-Advanced (LTE-A)** as an evolutionary enhancement designed to satisfy the strict architectural framework set by the International Telecommunication Union (ITU) under the **IMT-Advanced** umbrella [1].

The core objective of the LTE-Advanced architecture is to provide a highly reliable, packet-optimized structural flow. To achieve this, the architecture is built upon a *flat, all-IP* design philosophy that fundamentally differs from its UMTS/HSPA predecessor. The Radio Network Controller (RNC), which previously acted as a centralised bottleneck managing multiple base stations, has been entirely eliminated.

Intelligence is instead distributed directly to the **evolved NodeBs (eNodeBs)**, which autonomously manage radio resources, execute handover decisions, and communicate peer-to-peer via the **X2 interface**. The resulting architecture yields structural efficiency, improved network scalability, and simplified management [2].

The all-IP core network, called the **Evolved Packet Core (EPC)**, is responsible for session management, mobility anchoring, security, quality-of-service (QoS) enforcement, charging, and gateway connectivity to external packet data networks. The EPC rigorously separates the *control plane*—handled by the Mobility Management Entity (MME)—from the *user plane*—carried by the Serving Gateway (S-GW) and PDN Gateway (P-GW). This separation enables each component to be scaled independently, a design principle that directly informed the subsequent 5G Service-Based Architecture (SBA) [2].

This report provides a structural analysis of the LTE-Advanced system architecture, organized as follows: Section 2 presents the overall Evolved Packet System (EPS) structure and design principles; Section 3 details the User Equipment (UE) and its internal components; Section 4 examines the E-UTRAN radio access network; Section 5 dissects each core EPC node; Section 6 describes the protocol stack across both user and control planes; and Section 7 draws architectural conclusions.

2. LTE-Advanced General Architecture

The LTE-Advanced system is formally designated the **Evolved Packet System (EPS)** in 3GPP standards and is comprised of three principal domains [2]:

1. **User Equipment (UE)** — the subscriber terminal, encompassing both the physical hardware and the subscriber identity module.
2. **Evolved UTRAN (E-UTRAN)** — the distributed radio access network, composed of interconnected eNodeBs.
3. **Evolved Packet Core (EPC)** — the all-IP core network providing session management, mobility, security, and connectivity to external packet data networks.

As illustrated in Figure 1, the overall EPS architecture demonstrates how these three domains interconnect through a set of well-defined standardized interfaces, all carrying traffic exclusively over Internet Protocol (IP).

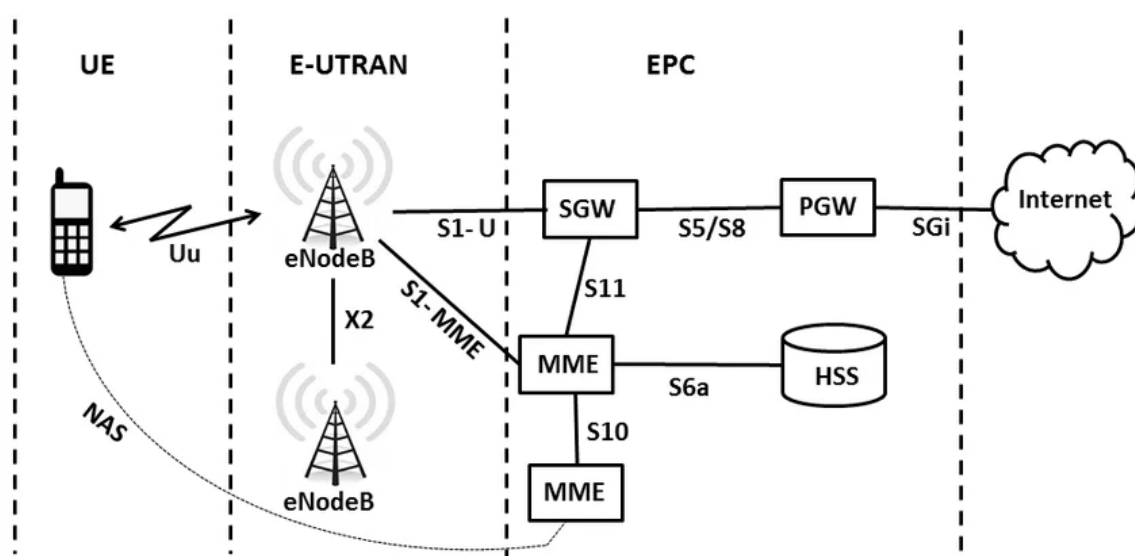


Figure 1: LTE-Advanced (EPS) Overall Architecture and Key Interfaces [4]

Table 2: Key Entities of the EPS Architecture and their Primary Functions

EPS Entity	Key Function
User Equipment (UE)	Provides the user interface and secure access to the network infrastructure.
E-UTRAN	Autonomously manages all radio resources and over-the-air communication.
Evolved Packet Core (EPC)	Handles core network signaling, IP data anchoring, and policy enforcement.

Based on Figure 1, the EPS architecture is governed by four fundamental design principles [2]:

- **All-IP backbone:** Every category of traffic—voice, data, and control signaling—is encapsulated and carried over IP, eliminating circuit-switched domains entirely and enabling unified transport.
- **Flat RAN without centralised RNC:** Each eNodeB independently manages all radio resources within its cell(s), executes Radio Resource Management (RRM), and performs handover decisions cooperatively with neighboring eNodeBs. This distribution removes single points of failure.
- **Strict Separation of Control and User Planes:** The MME exclusively handles control signaling (authentication, bearer management, mobility) and carries no user data whatsoever, while the S-GW and P-GW form the user-data forwarding path.
- **Distributed and direct handover:** eNodeBs communicate directly with each other via the X2 interface to execute handovers without routing radio-level signaling through the MME, reducing architectural latency compared to S1-based procedures.

3. User Equipment (UE)

The **User Equipment (UE)** is the subscriber-side terminal through which end users access the network architecture. The UE is composed of two logically and functionally separable components: the **Mobile Equipment (ME)** and the **Universal Subscriber Identity Module (USIM)**. This separation is architecturally significant—it allows a subscriber to use their service profile (USIM) in any compliant device (ME), and enables the network to independently identify the physical device versus the subscriber.

3.1 Mobile Equipment (ME)

The **Mobile Equipment (ME)** constitutes the physical hardware of the terminal. Its structural responsibilities span the complete LTE radio protocol stack and system-level functions:

- Implementing the full LTE radio protocol stack: PHY, MAC, RLC, PDCP, and RRC layers.
- Managing the air interface (Uu) with the serving eNodeB using **OFDMA** on the downlink and **SC-FDMA** on the uplink to maintain a low Peak-to-Average Power Ratio (PAPR).
- Handling **Non-Access Stratum (NAS)** signaling for session management, bearer establishment, and mobility procedures with the MME, transparently tunneled through the E-UTRAN.
- Executing power control, continuous radio measurements, and measurement reporting to support handover decisions at the serving eNodeB.
- Supporting advanced **MIMO** antenna configurations, enabling spatial multiplexing and diversity combining for enhanced signal robustness.

The ME is uniquely identified on the network by its **International Mobile Equipment Identity (IMEI)**.

3.2 Universal Subscriber Identity Module (USIM)

The **USIM** is a software application that resides on the **Universal Integrated Circuit Card (UICC)**. Its core function is to securely bind a subscriber's identity and credentials to the network session [2]:

- **Subscriber identity:** stores and presents the **IMSI** (International Mobile Subscriber Identity), a globally unique identifier.
- **Mutual authentication:** executes the **AKA (Authentication and Key Agreement)** protocol with the HSS. The USIM generates session keys used to encrypt and integrity-protect radio bearers and NAS signaling.
- **Profile storage:** maintains network-specific parameters and routing identifiers.
- **Security credential isolation:** the root secret key is permanently embedded in the USIM hardware and never leaves the USIM.

4. Radio Access Network (E-UTRAN)

The **Evolved Universal Terrestrial Radio Access Network (E-UTRAN)** constitutes the radio access domain of LTE-A. The defining architectural feature of E-UTRAN is the *complete absence of a centralised Radio Network Controller (RNC)*. As shown in Figure 2, every radio-layer function, from scheduling and link adaptation to handover execution, is performed autonomously by the **eNodeBs**.

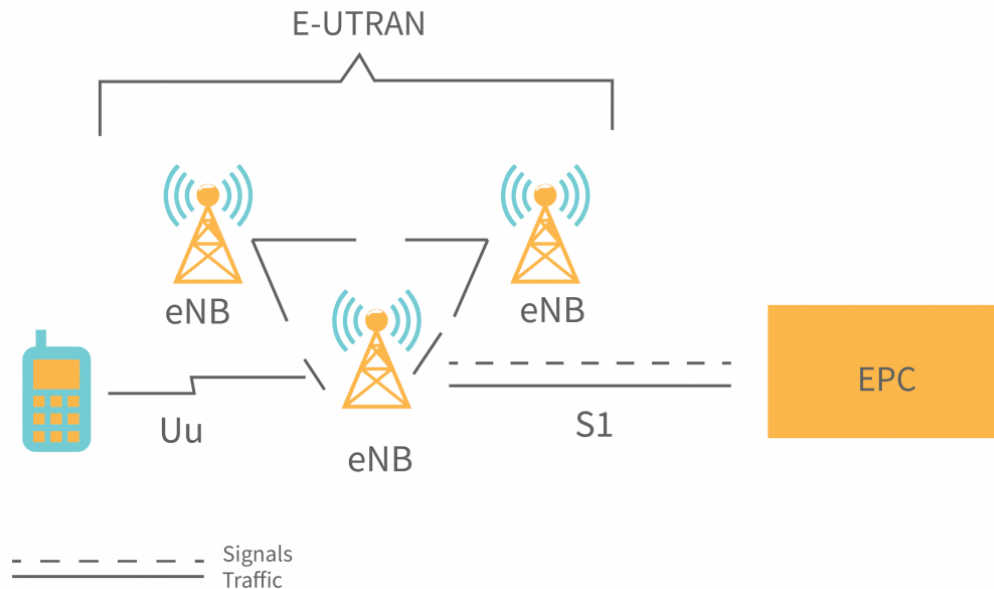


Figure 2: Internal Structure of the E-UTRAN and Interfaces with EPC and ME [3]

4.1 Evolved NodeB (eNodeB)

The **eNodeB (eNB)** is the fundamental network node of the architecture, acting simultaneously as the base station transceiver and as an intelligent radio resource controller. It absorbs all structural functions previously split between the Node-B and the RNC in UMTS [2].

Radio Functions:

- Air interface transmission and reception using OFDMA (downlink) and SC-FDMA (uplink).
- Comprehensive **Radio Resource Management (RRM)**: dynamic packet scheduling, adaptive link adaptation, power control, and Hybrid ARQ (HARQ) retransmission management.
- Advanced **MIMO** processing including spatial multiplexing and transmit diversity.
- Measurement configuration and UE reporting management for mobility and link quality monitoring.
- Ciphering and integrity protection of user plane data and signaling at the PDCP layer.

Control Functions:

- **RRC (Radio Resource Control)** connection establishment, reconfiguration, maintenance, and release; handover decisions and execution.
- Inter-cell load balancing via the X2 interface.
- Paging message distribution to UEs in ECM-IDLE mode.
- NAS signaling transport: the eNB transparently relays NAS messages between the UE and the MME.
- **Self-Organizing Network (SON)** functions for automatic neighbor relations and physical cell identity configuration.

Interfaces: The eNodeB terminates four key structural interfaces:

- **Uu:** the LTE air interface to UE, carrying both user data and control signaling.
- **S1-MME:** the control-plane interface to the MME, utilizing the S1 Application Protocol (S1AP).
- **S1-U:** the user-plane interface to the S-GW, transporting IP packets in GTP-U tunnels.
- **X2:** the direct peer-to-peer interface between eNodeBs, supporting control signaling and user-plane data forwarding during handovers.

5. Evolved Packet Core (EPC)

The **Evolved Packet Core (EPC)** is the all-IP core network. It provides a unified, packet-switched infrastructure for session management, mobility management, quality-of-service enforcement, security, and interworking with external packet data networks.

As presented in Figure 3, the EPC enforces a strict architectural separation between the *control plane* (MME, HSS, PCRF) and the *user plane* (S-GW, P-GW), allowing each structural path to scale independently.

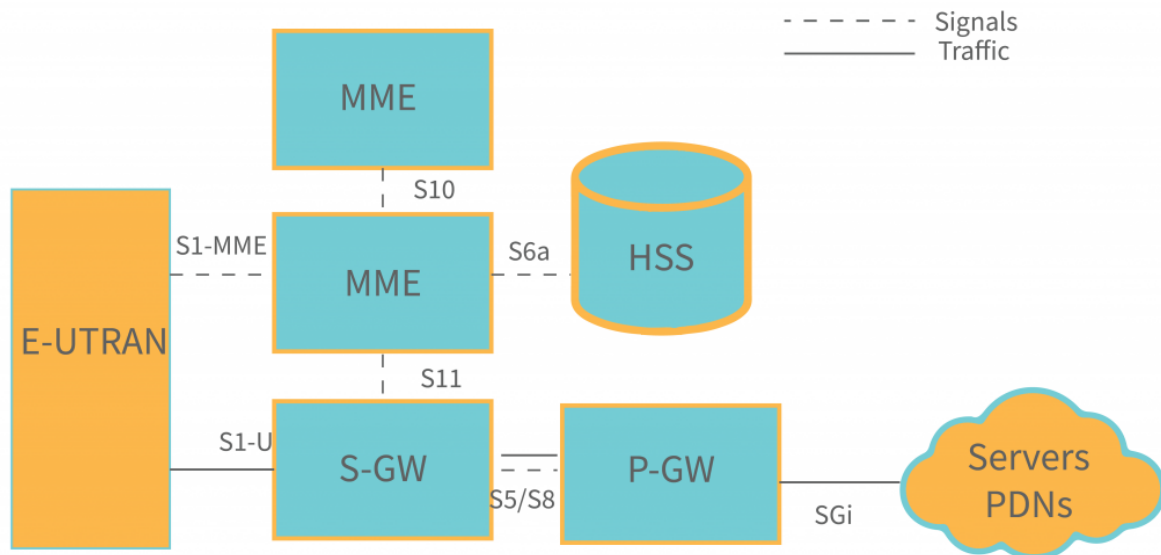


Figure 3: Evolved Packet Core (EPC) Detailed Architecture with Interfaces [3]

5.1 Mobility Management Entity (MME)

The **MME** is the principal *control-plane* node of the EPC. It handles all signaling between the UE and the core network but carries no user data whatsoever.

Key Functions:

- **Authentication and security:** initiates and controls the AKA procedure with the HSS and distributes NAS-layer security keys.
- **Session management:** manages the full lifecycle of UE registration, attach/detach requests, and Tracking Area Updates (TAU).
- **EPS Bearer management:** establishes, modifies, and releases default and dedicated bearers based on QoS architectural policies.
- **Mobility coordination:** manages S1-based handovers by forwarding context to target MMEs and prepares X2-based handovers.
- **NAS protocol termination:** acts as the endpoint for all NAS signaling from the UE.

5.2 Home Subscriber Server (HSS)

The **HSS** is the centralized *database and authentication center* of the EPC architecture [2].

Key Functions:

- Stores comprehensive subscriber profiles, routing identifiers, and roaming restrictions.
- Generates **authentication vectors** for the AKA protocol based on the shared root secret key.
- Maintains the address of the currently serving MME for each registered subscriber to enable incoming session routing.
- Communicates with the MME exclusively via the **S6a interface** using the Diameter protocol.

5.3 Serving Gateway (S-GW)

The **S-GW** is the *user-plane mobility anchor* for the E-UTRAN side of the EPC.

Key Functions:

- Routes and forwards IP packets between the E-UTRAN and the P-GW using **GTP-U tunnels**.
- Acts as the **mobility anchor for intra-LTE handovers**: user-plane tunnels are switched at the S-GW from the source eNB path to the target eNB path without re-establishing the connection to the P-GW.
- Buffers downlink data packets destined for UEs in idle state, triggering the paging procedure via the MME.
- Interfaces with legacy architectural nodes via the **S4 interface** for inter-RAT handovers.

5.4 PDN Gateway (P-GW)

The **P-GW** is the *IP-anchor point* between the EPC and external **Packet Data Networks (PDNs)**. It is the outermost gateway of the mobile network.

Key Functions:

- **IP address allocation**: assigns the UE's IP address from the APN's address pool; the address persists across handovers.
- **Policy and charging enforcement (PCEF)**: implements the PCC rules received from the PCRF, performing packet classification, traffic shaping, and gating control.
- **Deep Packet Inspection (DPI)**: classifies application flows for differentiated policy treatment.
- **SGi interface termination**: acts as the external IP interface toward the Internet or external packet networks.

5.5 Policy and Charging Rules Function (PCRF)

The **PCRF** is the *policy intelligence engine* of the EPC [2].

Key Functions:

- Generates and distributes **Policy and Charging Control (PCC) rules** to the P-GW to define the treatment of each EPS bearer.
- Dynamically updates rules via the **Gx interface** to enable real-time adaptation without session interruption.
- Translates application-layer requirements (received via the Rx interface from external session controllers) into concrete EPS bearer QoS parameters.

6. LTE-Advanced Protocol Stack

The protocol architecture is systematically partitioned into two major signaling strata [2]:

1. **Access Stratum (AS):** encompasses protocols terminated within the E-UTRAN (at the eNB and UE), including PHY, MAC, RLC, PDCP, and RRC.
2. **Non-Access Stratum (NAS):** operates between the UE and the EPC (MME) independently of the underlying radio technology.

6.1 User Plane Protocol Layers

Four protocol layers operate in the user plane within the eNB and UE to transport data payloads:

a. PHY — Physical Layer

The **Physical Layer** transforms data bits into radio signals. Key functions include:

- Generation of OFDM and SC-FDMA frames.
- **Modulation and coding:** adaptive selection of modulation schemes based on reported channel quality to maximize throughput.
- **MIMO processing:** codeword-to-layer mapping, precoding, and transmit beamforming.
- **Reference signal processing:** generation of signals for demodulation, channel estimation, and synchronization.

b. MAC — Medium Access Control

The **MAC layer** is responsible for radio resource allocation:

- Multiplexing and demultiplexing of RLC Protocol Data Units (PDUs).
- Dynamic packet **scheduling:** allocating Physical Resource Blocks (PRBs) to jointly optimize spectral efficiency and logical channel priorities.
- **Hybrid ARQ (HARQ):** error correction via chase combining or incremental redundancy retransmissions.

c. RLC — Radio Link Control

The **RLC layer** provides reliable data delivery:

- Operates in Acknowledged Mode (AM) for high reliability, Unacknowledged Mode (UM) for real-time media, or Transparent Mode (TM).
- SDU segmentation, re-segmentation, and concatenation to fill MAC transport blocks.
- In-sequence delivery and duplicate detection.

d. PDCP — Packet Data Convergence Protocol

The **PDCP layer** bridges the radio interface to the IP-based user plane:

- **IP header compression** using **ROHC** to reduce overhead for small packets like real-time voice payloads.
- **Ciphering and integrity protection** of user-plane data and control-plane signaling.
- **Duplicate detection** and reordering of PDCP SDUs during handovers.

6.2 Control Plane: RRC Layer

The **Radio Resource Control (RRC)** layer is the apex of the Access Stratum control plane. It operates between the UE and the eNB:

- **System information broadcast:** broadcasts configuration parameters.
- **Connection management:** manages establishment, reconfiguration, and release of RRC connections.
- **Mobility management:** configures measurement gaps, processes measurement reports, and executes handovers.
- **NAS message delivery:** acts as a transparent conduit for NAS signaling messages between the UE and the MME.

6.3 Non-Access Stratum (NAS)

The **NAS** layer operates entirely within the control plane between the **UE** and the **MME**. It is responsible for:

- **Session management:** initiating PDN connections and activating EPS bearers.
- **Authentication:** orchestrating the AKA procedure and generating security keys.
- **Mobility registration:** executing Tracking Area Updates (TAU) to ensure the MME knows the UE's location for paging.

7. Conclusion

This report has presented a structural analysis of the **LTE-Advanced architecture**, standardized by 3GPP as the *Evolved Packet System (EPS)*. The EPS represents a fundamental redesign of cellular network architecture, establishing the structural framework that underpins current deployments and the evolution toward subsequent network architectures [1, 2].

The key architectural insights can be summarized as follows:

1. **Flat, all-IP design:** The elimination of the centralized RNC and the adoption of a fully IP-based transport backbone yield higher scalability through distributed intelligence and drastically simplified network operations.
2. **Distributed radio intelligence:** The eNodeB consolidates scheduling, link adaptation, and handover execution into a single node, coordinating with peers via the X2 interface without core network involvement for latency-critical procedures.
3. **Strict control/user plane separation:** The MME carries only signaling, while the S-GW and P-GW carry only data. This structural separation enables each component to scale and be virtualized independently.
4. **Layered protocol stack:** The user-plane stack (PHY / MAC / RLC / PDCP) and the RRC and NAS control layers each address a clearly scoped set of functions, enabling independent optimization.
5. **Policy framework:** The PCRF-driven PCC framework enables fine-grained, per-bearer policy enforcement from the radio access network to the external packet data gateway, guaranteeing strict delay requirements, differentiated throughput, and prioritized routing.

By enforcing strict separation of planes, distributing radio control to the edge, and unifying all transport over IP, the LTE-Advanced architecture established the foundational structural principles that continue to guide the development of modern mobile telecommunications systems.

References

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